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ABSTRACT

Background and Aims: Alzheimer's disease (AD) is a progressive neurodegenerative disorder of aging characterized by memory loss and cognitive decline, associated with amyloid- β toxicity, tau hyperphosphorylation, and neurofibrillary tangle formation. Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) affects the central nervous system and has been linked to neurological manifestations and accelerated neurodegeneration, including in individuals without pre-existing AD. Emerging evidence suggests COVID-19 may increase levels of hyperphosphorylated tau, potentially worsening AD severity. This review synthesizes current knowledge on COVID-19–AD interactions and evaluates proteolysis-targeting chimeras (PROTACs) as an emerging therapeutic strategy.

Methods: Narrative synthesis of recent literature on SARS-CoV-2–related neuropathology, tau pathology in AD, and the design and preclinical development of PROTACs targeting disease-relevant proteins via the ubiquitin–proteasome system.

Results: Reports indicate COVID-19 can precipitate or exacerbate neurodegenerative processes and is associated with increased tau phosphorylation and other biomarkers of neuronal injury. Conventional AD therapies remain limited in efficacy. PROTACs—heterobifunctional molecules that recruit target proteins to E3 ligases for proteasomal degradation—do not require classical active-site binding and have demonstrated preclinical potential for degrading pathogenic proteins, supporting their exploration against tau-driven pathology in AD.

Conclusions: COVID-19 may intensify AD pathogenesis through mechanisms that include tau hyperphosphorylation, underscoring the need for targeted interventions. PROTACs offer a mechanistically distinct, protein-degradation–based approach with promise for modifying tau-mediated disease; rigorous preclinical and clinical studies are warranted to establish feasibility, safety, and therapeutic impact in AD.

1 | Understanding Alzheimer's Disease: An

Introduction

Alzheimer's is a neurodegenerative disease which leads to decline in learning, memory, behavior, and cognitive functions of the diseased individual. The Hippocampus region within the brain, responsible for maintaining learning and memory, is affected by deposition of β -amyloid or tangling of neurofibrils

[1]. Symptoms of Alzheimer's disease generally appear in patients who are more than 60 years old. Figure 1 depicts how the brain of a diseased person differs from that of a non-diseased person. There is marked shrinkage in the areas that involve learning, memory and other cognitive functions.

Earlier therapies for Alzheimer's disease were targeted towards β -amyloid. These included $\beta\gamma$ secretase inhibitors which

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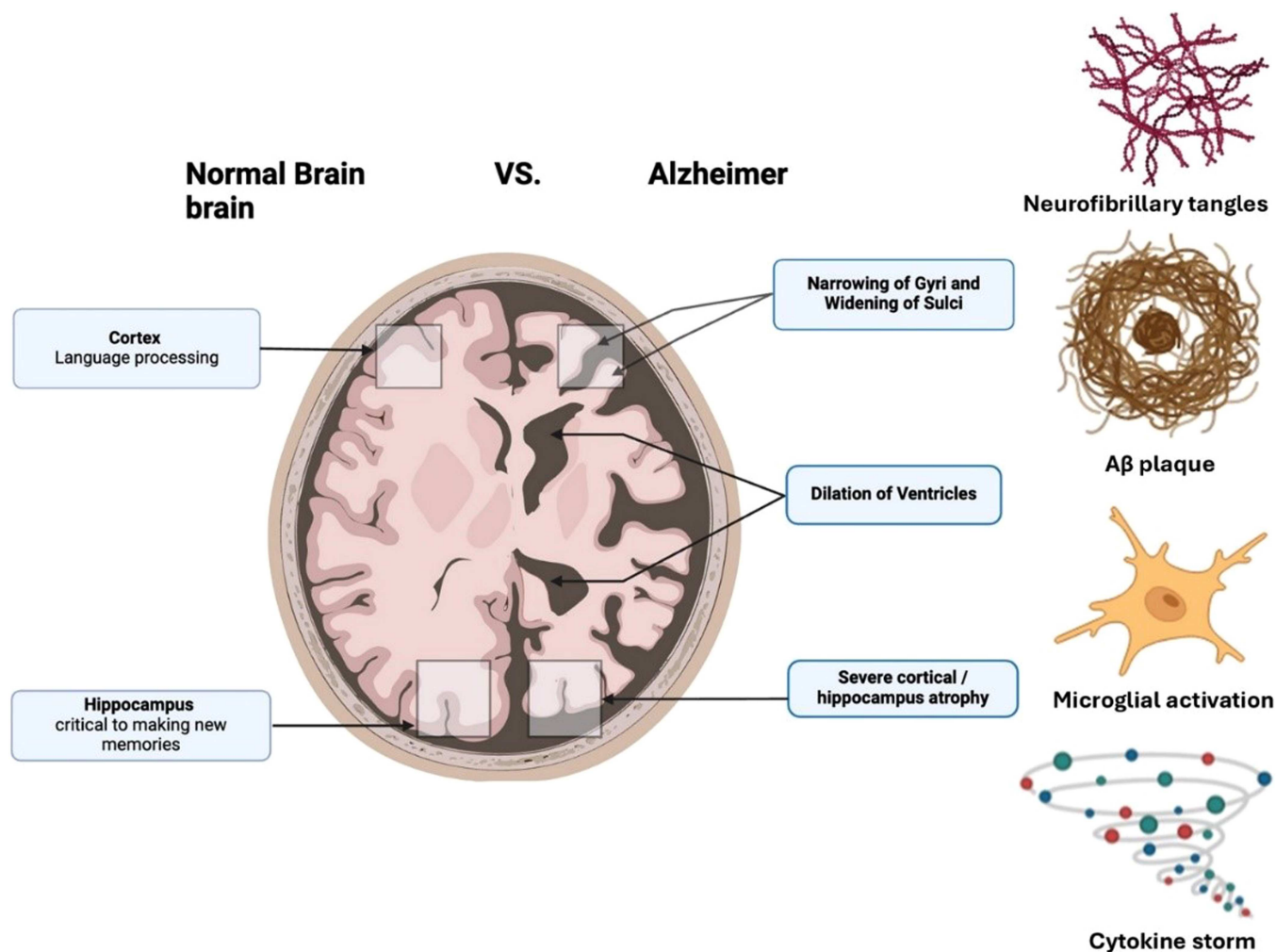


FIGURE 1 | Comparison of normal human brain and Alzheimer's brain.

reduced production of β -amyloid [2] or the removal of β -amyloid by facilitating its degradation or producing antibodies against them [3, 4]. However, these therapies did not have significant results. The challenge lies in accessing the proper regions in the brain and thereby possible side effects when reaching non-target tissues. The blood–brain barrier is the major hindrance for drugs targeting the brain cells [5].

In the recent times, the Tau protein has been targeted for therapy. The approaches mainly focus on reducing its expression by generating small interfering RNA molecules or anti-sense mRNAs [6], regulating enzymes involved in the post-translational modification of Tau, production of inhibitors which aggregate Tau [7]. Even harboring the ubiquitin-proteasome system or autophagy-lysosome pathway for degradation of Tau protein can also be a probable approach [8].

Studies have indicated that Alzheimer's disease is inversely related to cancer. This is an assumption done on the basis of the biological pathways followed by these diseases. Age being one of the major contributors both in the case of cancer and Alzheimer's disease, studies indicated that pathogenesis of either one of them protects patients against the other disease [9]. However, the reasons behind such protection are still not known. Cell cycle re-entry is required by both cancer and Alzheimer's disease pathogenesis [10]. Apart from this, there are many more common pathological features. In case of

Alzheimer's disease, neural cell cycle is stopped instead of further cell division. It includes a substantial increase in CDK2, CDK4, CDK5 and activation of caspase for amyloid precursor protein phosphorylation, proteolysis and generation of amyloid β during cell cycle. mTOR is involved in the pathophysiology and neural cell cycle re-entry of Alzheimer's disease, and therefore it is activated at the beginning of proliferation points. mTOR activates Tau which results in the building up of neurofibrillary tangles and causes neurodegeneration. Furthermore, the process of autophagy is activated for the removal of amyloid β from cells when the mTOR pathway is deactivated [11].

This review summarises the potential impact of COVID-19 on patients who were previously suffering from Alzheimer's disease. It also provides a critical link between the viral mechanism and the neurodegenerative changes occurring. Finally, a possible role of PROTACs as treatment strategy is also discussed.

2 | The Interlink Between Alzheimer's and COVID-19

COVID-19 has been associated with neurotropic symptoms like headache, stroke, and anosmia thereby affecting the central nervous system. Aged people are more sensitive to viral

infections which is further aggravated if there is impairing of cognitive functions due to old age, pre-existing morbidity, and low immunity. SARS-CoV-2 consists of a lipid-like envelope, which is obtained from the cell membrane of the host, to enclose the genetic material and other proteins of the virus. The entry of the virus into the host cell is facilitated by the spike protein S which binds to the ACE-2 (angiotensin converting enzyme) receptor on the surface of the host cell, mainly the ciliated epithelial cells in the airway passage [12]. The ACE-2 receptor is also found in the intestinal and kidney epithelial cells, and expressed in the brain as well [13]. In the brain, ACE-2 receptors are expressed by the cells comprising the blood-brain barrier, the pericytes and astrocytes, which are targeted by the coronavirus to enter into the Central Nervous System [14]. The definitive evidence of sustained CNS infection in humans still remain limited.

Alzheimer's disease as well as old age decreases the integrity of the blood-brain barrier [15]. However, the coronavirus has also been found to exploit the olfactory bulb to enter the central nervous system directly thereby bypassing the blood-brain barrier which can be proven by the existence of ACE-2 receptors in the olfactory bulb of mice and humans [16]. It has also been hypothesized that there may be overexpression of ACE-2 receptors in brains affected by Alzheimer's disease, indicating that patients suffering from Alzheimer's may become more vulnerable on infection with COVID-19. There is overexpression of the amyloid- β precursor protein in the olfactory bulb [17]. The deposition of amyloid- β precursor protein in the olfactory bulb has also been found in the early stages of Alzheimer's in humans before the onset of symptoms [18]. Alzheimer's disease further affects the functioning of the olfactory bulb, making it further susceptible to viral entry. The spike protein S of SARS-CoV-2 binds to the ACE-2 receptor, thereby downregulating its expression and in turn, increases the expression of Angiotensin II [19]. This causes vasoconstriction in the brain which leads to hypoxia or hypoperfusion as seen in cognitively impaired patients post Covid [20]. This may occur due to the oxidative stress imposed by formation of reactive oxygen species by NADPH oxidase which was activated by Angiotensin II [21]. This oxidative stress, inflammation and hypoperfusion in the brain may altogether lead to neurodegeneration. Also, as indicated in research, with downregulating expression of ACE-2, the spike protein has also been found to upregulate the expression of amyloid- β precursor protein which progresses into Alzheimer's disease [22].

Alzheimer's disease was found to be the most common disease in patients who suffered from COVID-19 [23]. Age is the significant factor in case of Late Onset Alzheimer's disease (LOAD) as most of the cases are seen to develop in patients who are above after 65 years [24]. The genetic component responsible for such risk is the gene coding for apolipoprotein E, commonly called APOE. APOE in humans have three alleles namely, $\epsilon 3$, $\epsilon 2$, and $\epsilon 4$ out of which the former two are more protective and associate with less risk while the latter has increased risk and raises the probability of Alzheimer's by three fold. In homozygous conditions, that is, when there are two copies of $\epsilon 4$ (APOE4), there is an eight-fold increased risk of Alzheimer's and also twice as many chances of the patient suffering from coronavirus infection and also facing fatality by COVID.

It has been seen that higher levels of Tau protein in cerebrospinal fluid are seen in females compared to males and

women have been found to be more common carriers of APOE4 alleles. This point also suggests that APOE4 may cause neurodegeneration within a particular sex [25]. Also, post infection, there is more mental fatigue in APOE4 carriers, and the possible reason could be cerebrovascular dysfunctionality [26]. Figure 2 thereby illustrates the possible interlinkage between Covid-19 and Alzheimer's disease in aged people and the APOE4 allele affects it.

3 | Role of APOE4, Amyloid- β , and Tau in COVID-19

In vivo experiments on mouse models have also shown higher viral infection and higher death rates as well as weakened immune system in mice in which APOE4 gene has been inserted [27]. APOE4 increases permeability of the blood-brain barrier and reduces its integrity, which facilitates entry of virus into the central nervous system.

Studies have also shown that APOE4 promotes COVID infection by increasing levels of cholesterol, thereby regulating cholesterol homeostasis which also promotes entry of virus by facilitating the interaction between viral spike protein, S with the ACE-2 receptor [28] as shown in Figure 3. APOE4 also leads to excess release of inflammatory cytokines which causes a cytokine storm which increases severity of COVID-19 infection [29]. The APOE4 mouse model, although viable, has limitations as mice physiology, immune responses and viral tropism clearly differ from that of humans. Thus, complete replication of complex, genetic, environmental and clinical features of COVID-19 in APOE4 carriers is not possible.

SARS-CoV-2 disturbs amyloid- β -42 and Tau protein homeostasis. In vitro experiments have shown that organoids infected with coronavirus have higher levels of hyperphosphorylated tau and their distribution patterns are also altered [30]. Coronavirus induces toxicity of β -amyloid in the brain and inhibits its clearance in serum of blood, thereby leading to its aggregation in the cerebrospinal fluid [31]. Amyloid- β -42 also facilitates binding of spike protein S1 to the ACE-2 receptor, thereby promoting its entry, thus leading to excess production of inflammatory cytokines like IL-6 which causes impairment of hippocampal neurons [32]. While these findings raise concern for accelerated Alzheimer's disease pathology, causality in humans has not been firmly established. They are still largely preclinical.

Post-mortem reports of patients who suffered from SARS-CoV-2 infection showed accumulation of immune cells and gliosis, which caused leakage and disruption of blood-brain barrier and damaged the axons [33]. Astrocytes in the blood-brain barrier, which are already susceptible to coronavirus infection, also promote the secretion of inflammatory cytokines and increase production of amyloid- β [34]. These detrimental occurrences may ultimately lead to Alzheimer's disease [35].

4 | Inflammatory Response in Case of COVID-19 and Alzheimer's Disease

In Alzheimer's disease, there are increased levels of inflammatory molecules in the brain caused by extra neuronal

deposits of amyloid- β , damaged neurons or tangling of neurofibrils. These inflammatory molecules accumulate for a long time which ultimately damages the neurons [17]. This also supports the fact that Alzheimer's disease establishes in an individual before symptoms are seen. There are two receptors associated with inflammation which increase the severity of Alzheimer's. One is TREM2 protein, which is the triggering receptor found on myeloid cells. This TREM2 activates immune cells like microglial cells, dendritic cells, and macrophages [17]. There is higher expression of TREM2 in the peripheral blood of patients suffering from Alzheimer's [36]. The other inflammatory receptor is CD33 which attenuates immune response by mediating cellular interactions and increases susceptibility of individuals to Alzheimer's [37]. It is overexpressed in the microglial cells of the brain of Alzheimer's patients [38].

It has been found that in the T cells of lungs and peripheral blood of COVID-19 patients, expression of TREM2 protein is increased [39]. Also, the spike protein of coronavirus is an appropriate ligand of CD33 receptor which on binding, increases the response of suppressor cells derived from myeloid as well as monocytes which increases seriousness of infection [17]. Thus, Alzheimer's patients, having single nucleotide polymorphism in CD33 or other variants of TREM2, are likely to suffer more from COVID-19. COVID-19 complications have also been prolonged by the action of inflammatory molecules. The virus causes activation of microglial cells which regulate the expression of cytokines like tumor necrosis factor-TNF- α , interleukins like IL-1 β and IL-6, ultimately aggravating neurodegeneration and increasing dementia [40].

Inflammasome is another molecule, belonging to the family of NOD immune system receptors which causes crosstalk between Alzheimer's disease and COVID-19. Especially, NOD-like receptor Pyrin domain NLRP-3 inflammasome has been associated with inflammation caused by COVID-19 [41]. NLRP-3 impairs the phagocytic activity of microglial cells which is associated with clearance of amyloid- β [42]. It also promotes Tau protein expression which causes the progression of Alzheimer's [43]. Figure 4 illustrates the possible mechanistic link between COVID-19 and Alzheimer's disease.

5 | PROTACs Mediated Protein Degradation

PROTACs or proteolysis targeting chimeras are molecules with heterobifunctional properties which bind to two or more molecules at the same time. They consist of three domains—the moiety which binds the target protein (POI), a linker domain and a domain which binds E3 ubiquitin ligase [44]. PROTAC binds to the ligase and POI, thus forming the PROTAC-POI-E3 complex which is a ternary structure [45]. The ubiquitin proteasome system (UPS) is responsible for maintaining homeostasis in eukaryotic cells by degrading and eliminating damaged and deformed proteins [46]. This is done by substrate specificity. The UPS degrades protein via three main enzymatic steps. E1 activates free ubiquitin which is transferred to E2 by thio-transesterification. E3 catalyzes substrate specific ubiquitination thereby targeting proteins for 26S proteasome-mediated degradation [47]. At the same time, PROTACs also bind E3 and the target protein, thus bringing them within spatial distance, facilitating specific substrate recognition by E3 ligase.

Therefore, PROTACs mediate protein degradation in cells by hijacking the ubiquitin-proteasome system, as shown in Figure 5 [48].

Usually, drug designing is done by recruiting small molecules which bind to target proteins with high affinity, occupy their active site and thus inhibit their function [49]. However, there are some target proteins which lack such binding or active sites for binding of drug molecules. PROTACs make it possible to degrade these proteins which are otherwise considered undruggable, because their function does not include binding to active sites. Thus, they increase the spectrum of target proteins they can degrade. For example, androgen and estrogen receptors have been targeted by PROTACs for degradation and this method has proceeded into secondary clinical trials already [50, 51]. PROTACs decrease enzymatic and non-enzymatic activities of proteins, substrate specificity, and overpowering drug resistance. Also, less dosages of PROTACs are required than traditional drugs to degrade target proteins [52]. The non-requirement of active sites, make PROTACs a potential therapeutic in case of Alzheimer's disease, where tau lacks easily druggable active sites. Some of the new age therapeutics use PROTACs that degrade oncogenic proteins selectively by utilizing the E3 ligases through the ubiquitin-proteasome pathway [53]. Thereby, PROTACs can be a promising new approach for precision-targeted therapy in case of cancers of lung and ovary [54].

6 | Role of PROTACs in Neurodegenerative Diseases

Neurodegenerative diseases occur in elderly people due to the accumulation of misfolded proteins such as Tau, β -amyloid, α -synuclein which form aggregates [55]. Tau is a protein associated with microtubules for maintaining its stability and regulates transport through axons. Aggregation of Tau in the brain, affects the β -amyloid toxicity, and ultimately leads to Alzheimer's disease [56]. Hyperphosphorylation of Tau protein causes the tangling of neurofibrils which ultimately aggregate, leading to Alzheimer's.

Alzheimer's disease can be treated by controlling the aggregation of Tau. A peptide-based PROTAC (PROTAC12) had been designed which consisted of Keap1 E3 ligase, having high affinity for Keap1 and Tau protein and leading to their immunoprecipitation. It moderately degrades Tau protein. Figure 6 illustrates the possible PROTAC-mediated degradation of Tau via the ubiquitin-proteasome pathway, where Keap1-E3 ligase facilitates ubiquitination and subsequent proteasomal degradation of Tau aggregates [57].

PROTAC11 is another synthetic molecule which shows the highest degradation potential of Tau protein. It is able to pass through cell membranes and promote polyubiquitination of Tau which in turn reduces β -amyloid cytotoxicity. Another molecule, PROTAC13 is found to degrade both wild type and mutant forms of Tau protein in neurons associated with frontotemporal dementia. It has the ability to penetrate the blood-brain barrier. PROTAC14 shows degradation of Tau in both healthy and diseased conditions by in vivo and in vitro analysis. A single dose of this PROTAC14 significantly decreases Tau levels in the brain which also reduced toxicity brought about by β -amyloid,

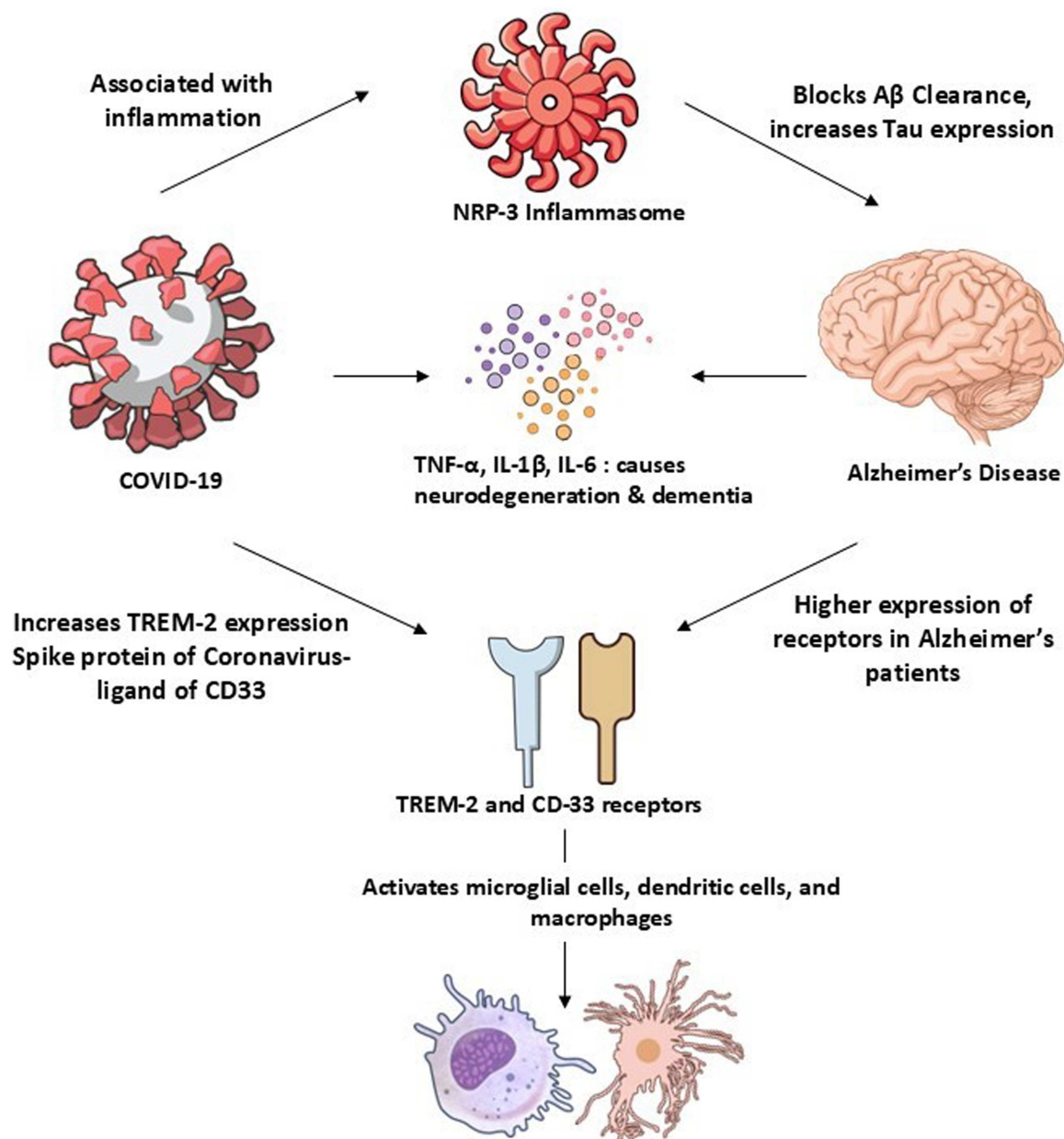


FIGURE 4 | Inflammatory response in case of COVID-19 and Alzheimer's disease. SARS-CoV-2 infection activates the NLRP3 inflammasome and elevates pro-inflammatory cytokines (TNF- α , IL-1 β , IL-6), leading to neuroinflammation, impaired amyloid- β clearance, and increased tau expression. The viral spike protein also enhances TREM-2 and CD33 receptor activity, promoting microglial and macrophage activation.

similar to PROTAC11. This made PROTAC14 a promising drug for the treatment of Alzheimer's disease. Therefore, we can conclude them as a promising cure to neurodegenerative diseases in humans [58].

Another study revealed that PROTAC molecule, C004019, is able to proficiently degrade and clear Tau protein in the brain in HEK293 cells via the ubiquitin-proteasome system. The test was carried out in mice model and even functions at very low dosage. Though Tau has a longer half-life of 23 days [59], 50% of it reduced with a single C004019 treatment after 8 days. Also, infrequent dosage of C004019 every 6 days for a month showed positive results [28].

HEK293 cells which stably expressed wild-type full-length human Tau is referred to as HEK293-hTau. In hTau-

overexpressed mice, PROTAC C8 was seen to decrease levels of both total and phosphorylated Tau protein. The results obtained from Western blot shows that C8 broke down Tau protein in a time dependent fashion by the means of ubiquitin-proteasome system [60]. However, all these findings are mainly limited to preclinical settings and their clinical applications in case of Alzheimer's disease remain largely uncertain.

7 | GSK-3 β as a Possible Drug Target

Glycogen synthase kinase 3 β (GSK-3 β), which is a serine/threonine kinase, is constitutively expressed. Apart from being linked to a number of illnesses, which includes diabetes, cancer, inflammatory, immunological, and

A variety of PROTACs has been designed which are used to increase GSK-3 β degradation in cells [64]. One study discussed the synthesis of a series of GSK3 degraders. A number of extremely strong GSK3 degraders are known. In the case of SH-SY5Y cell lines, a PROTAC, namely, PT-65 demonstrated the strongest degradation ability against GSK3 α and GSK3 β . The SPR assay further established the high binding affinity of PT-65 to GSK3 β . The proteome analysis disclosed that GSK3 degradation might be specifically activated by PT-65. Furthermore, PT-65 could safeguard SH-SY5Y cells from amyloid β -induced cell impairment and efficiently decrease GSK3 β and amyloid β -mediated Tau hyperphosphorylation in a dose-dependent manner. In addition, it was found that PT-65 may lessen Tau hyperphosphorylation brought on by OA and improve memory and learning shortfalls in an in vivo model of Alzheimer's Disease.

Another study combined the action of PROTAC and a nanoparticle delivery method, which can efficiently penetrate the blood-brain barrier, to produce a combined novel nanodrug called NPD. The nanodrug is prepared by using positively charged nanoparticles which are loaded by DNA intercalation with a cationic PROTAC, peptide 1 which targets Tau protein. This nanodrug showed high efficiency in encapsulation, was stable in serum, efficient drug release ability, and permeability through blood-brain barrier. In cell culture of neurons and brains of mice suffering from Alzheimer's, NPD drug potentially cleared Tau protein. The drug also enormously improved cognitive functions of diseased mice when intravenously injected in the mice. There were no significant abnormalities seen, proving it to be clinically beneficial. Thus, this nanomedicine shows high potential to be utilized to treat Alzheimer's where it decreases Tau levels in brain lesions and improves cognitive functions [65].

In conclusion, PROTACs represent an emerging therapeutic modality with disease modifying potential for Alzheimer's disease. However, their efficacy in mitigating post COVID neurodegeneration remains a well substantiated hypothesis requiring extensive preclinical and clinical validation.

8 | Future Prospects

PROTACs are now extensively used as a viable treatment option for diseases, which include immunological disorders, cardiovascular dysfunction, neurodegenerative disorders and also cancer hence, they are valued from both industrial and academic perspectives. Due to the complex tertiary structure of PROTAC, large masses of proteins are also degraded, which are otherwise not degraded by the barrel-shaped proteasome.

PROTACs have limited usage due to its susceptibility to protease degradation and less in vivo permeability. One of the PROTACs, C004019, is capable of crossing both the blood brain barrier and the proteasome, unlike most of the drugs, which improves the synaptic and cognitive functions in mice. However, due to the natively unfolded structure of Tau, the recognition ability of C004019 for Tau species is relatively poor [13].

In the treatment of Alzheimer's disease with PROTACs, to overcome these challenges and increase the stability many approaches have been undertaken. Among them, include

substituting unnatural amino acid residues, modification of structural backbones of the compounds, and cyclization [66]. To increase the permeability in vivo, hydrogen bonding can be decreased along with increasing lipophilicity [67]. The creation of an active PROTAC may need some more additional factors apart from having a high affinity towards the target protein. This activity can be enhanced by the induction of steric structures by creating target protein associated with hijacked E3 ligase and transfer of ubiquitin from E2 to target protein [68].

The development of new ligands must be one of the focuses for further development of PROTACs. Among the 600 E3 ligases that are present in the human genome, only a small number of them are currently employed in PROTAC technology. This indicates enormous potential to test and identify more such E3 ligases so that they can be used in PROTAC technology. Off-target reactions must be prevented so that prompt treatment can be carried out in case of Alzheimer's disease [69].

Three plasma membranes must be crossed by intravenously delivered PROTAC molecules for central nervous system disorders which comprise of apical and basolateral membranes of the capillary endothelial cells present at the blood-brain barrier and the plasma membrane of target cells. This is because the ubiquitin-proteasome system functions inside target brain cells. Canonical PROTAC compounds often show low membrane permeability and low cell selectivity. Cell selectivity is absent from passive diffusion across the membrane. Heparan sulphate proteoglycans (HSPGs) are widely expressed on the surface of several cell types. Therefore, it is significant to carry out highly selective internalization into target cells, especially the capillary endothelial cells at the blood brain barrier, in order to prevent off-target antagonistic effects. At the blood-brain barrier, two strategies are used which include receptor-mediated transcytosis by a transferrin and insulin receptors and carrier-mediated transport. The carrier-mediated transport is carried out by proton-coupled organic cation antiporter. These are non-invasive trans-endothelium approaches. A transferrin receptor is frequently engaged for carrying out receptor-mediated endocytosis. Ligand-receptor complexes are endocytosed when a blood ligand attaches itself to a transferrin receptor present on the apical membrane of the capillary endothelial cells. Based on the union of the endosome and the basolateral membrane through the secretory pathway rather than the degradation pathway leading to lysosomal degradation, a ligand is released from the receptor in an endosome as acidification because of endosomal development. Its exocytosis is then carried out to the brain parenchyma through the brain interstitial fluid (ISF). As a ligand, anti-transferrin receptor monoclonal antibodies are used. Through receptor-mediated transcytosis, well-designed antibodies, such as antibody-PROTAC conjugates or antibody-encapsulated nanoparticles, can decisively breach the blood-brain barrier [70].

PROTACs hold immense potential as a possible treatment of neurodegenerative diseases like Alzheimer's by targeted degradation of accumulated proteins. However, challenges of stability, permeability and selectivity still persist. Future studies must be aimed to optimize these challenges.

Authors Contributions

R.B., T.B., and B.D. were involved in the conceptualization, proof-reading, formatting, illustrations, and preparation of the final